

Udemy ML, DL, NLP

course
Kruskal's Algorithm
Module 42 :-

DBSCAN
Clustering

lec 1 :- DBSCAN Functioning

lec 2 :- Examples

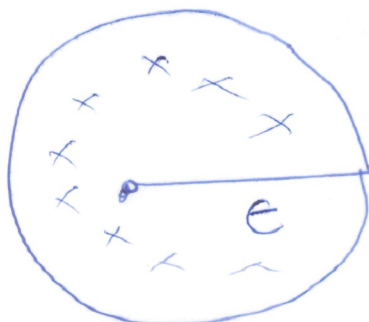
lec 3 :- Pros & Cons
non linear
clustering

- (Red Point) $\rightarrow$ Core Point
- (Yellow) $\rightarrow$ Border Point
- (Blue) $\rightarrow$ Outlier

min points = 4

ϵ = radius

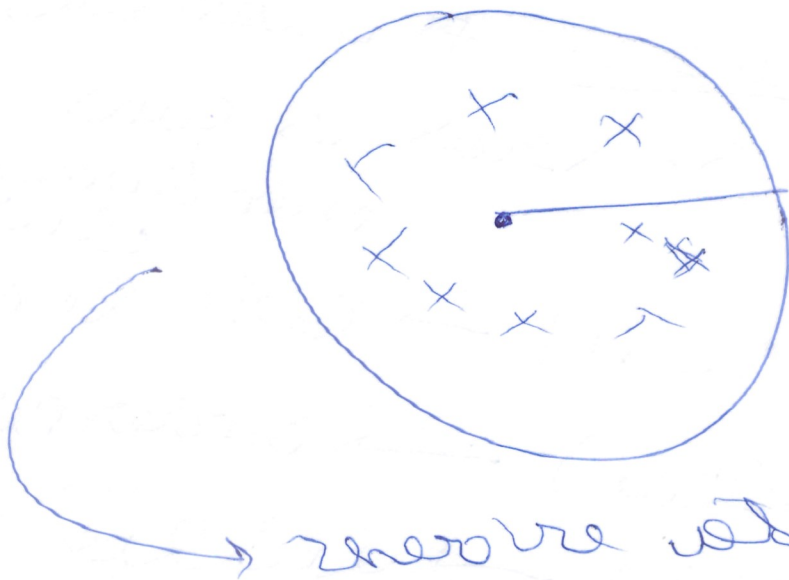
Core Point :-



• (Red Point)

③ Change for no. of points
that should exist = 4
min point = 4

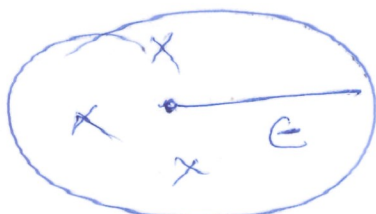
① No. of points within
the ϵ should be
greater ≥ 4



remove other data
points existing in space

Border Points:-

Change for no. of
points within the radius will
be less than minimum
change



Outlier :-



→ either also
call as noise
is a data
in the result

point that through
low density area. It does
not have enough neighbors.

~~For~~

⇒ Examples of applying
DBSCAN

DBSCAN can find non
linearly separable clusters.

The dataset cannot be easily
clustered with K-means

• DBSCAN groups data
points based on spatial
density. making it ideal
for real datasets.

Examples are as:-

(4)

- ① Crime Mapping
- ② Urban Planning
- ③ Earthquake Monitoring

⇒ Pros & Cons of DBSCAN

Advantages

- ① DBSCAN does not require one to specify the no. of clusters in advance.
- ② DBSCAN can find the arbitrary shaped clusters. It can even find a cluster completely.
- ③ DBSCAN has no notion of noise.

⑤ DBSCAN requires the 2 parameters ϵ is mostly insensitive to the ordering of points in database

④ DBSCAN is designed for use with databases that can accelerate region look up queries as - query on page

⇒ Disadvantages :-

① DBSCAN is not deterministic
border points that are not part of cluster can be a part of either cluster.

② The quality of DBSCAN depends on distance measure